Homework 2: Factor Models and Performance Measurement

Quantitative Methods in Investment Management · bs3759 · Sample: January 1994 to June 2012

Data and conventions. All results come from my notebook (bs3759_HM2.ipynb), run on the data file Performance_Measurement_Spreadsheet-1.xlsx. I use monthly returns in decimal units over the full sample. Excess return: $E_t = R_t - RF_t$. Active return: $A_t = R_t - R_t^b$, where R_t^b is the DJCS Hedge Fund Index. Annualization: arithmetic averages are multiplied by 12; the geometric return is $\left(\prod_{t=1}^{T}(1+R_t)\right)^{12/T} - 1$; volatility and tracking error are multiplied by $\sqrt{12}$ (sample standard deviation). Sharpe ratio $= 12\bar{E}/(\sigma_E\sqrt{12})$, information ratio $= 12\bar{A}/(\sigma_A\sqrt{12})$, appraisal ratio $= 12\hat{\alpha}/(\sigma_\varepsilon\sqrt{12})$. Cumulative wealth: $V_0 = 1$ (100 in the chart), $V_t = V_{t-1}(1+R_t)$; high-water mark $HWM_t = \max(V_0,\ldots,V_t)$; drawdown $DD_t = 1 - V_t/HWM_t$; maximum drawdown $= \max_t DD_t$. Skewness and excess kurtosis are pandas' sample estimators (0 for a normal distribution). Returns and volatilities are in percent, ratios in decimals.

Part I: Factor models and performance interpretation

1. Equilibrium model vs. time-series factor model

The equilibrium asset-pricing model (CAPM), $E[R_i] = r_f + \beta_{iM}(E[R_M] - r_f)$, investigates the sensitivity of an asset's return to the market and predicts that all of an asset's expected return should come from that market exposure, at a rate given by the market premium. It contains no leftover term. The time-series factor model, $R_{p,t} - r_{f,t} = \alpha + \beta'\mathbf{f}_t + \varepsilon_t$, breaks down a portfolio's realized return into $\beta'\mathbf{f}_t$ (the return coming from exposure to a factor), α (extra return not justified by that exposure), and residual noise. In the performance setting, the intercept is exactly α: the average excess return that is left after removing what the factor exposures explain. For this reason, a positive alpha alone cannot establish manager skill: it is heavily model-dependent and can shift meaningfully with a different factor set, it can change based on the sample window used, and it may not persist out of sample. A positive alpha is reason enough to investigate further, but not sufficient on its own to conclude a manager's skill.

2. Adding size, value and momentum

The manager's realized return has not changed: it is the same historical outcome regardless of which regression is used to explain it. What has changed is the attribution of that return. In the market-only regression, any return not explained by market exposure was assigned entirely to alpha. Once size, value and momentum factors are added, part of that same return is now correctly attributed to $\beta'\mathbf{f}_t$ for those additional factors, since the manager was apparently exposed to them. The shrinkage in α reveals that what initially looked like manager-specific skill was, at least in part, exposure to known systematic factors.

Part II: Data audit and sensitivity analysis

3. Data audit

Table 1: Data audit

Item	Value
Start date	31 January 1994
End date	29 June 2012
Frequency	Monthly (business month-end)
Total observations	222
Usable observations	222

In the spreadsheet, the first two rows contain the titles and the data sources, and the column names are in row 3. I loaded the file using row 3 as the header, named the first column "Date" (its label is in row 2), and dropped the completely empty columns (B, D and I) and any empty rows at the bottom. I also removed a leading space in the Convertible Arbitrage column name. After cleaning, the dataset has 222 monthly observations and 16 columns: the date, the risk-free rate, the four factors, the nine strategy indexes and the DJCS index. There are no missing values and no placeholder values (such as -99). All return columns are stored as numbers (`float64`). All return series are in decimal form (e.g. 0.02 = 2%): the mean absolute monthly values are between 0.25% and

3.9% and no maximum reaches 0.5, so no rescaling was needed.

4. Equity Market Neutral: largest absolute monthly return

Table 2: Equity Market Neutral: full sample vs. excluding the largest observation (November 2008)

	Full sample	Excl. largest obs.
Geometric average (monthly)	0.38%	0.62%
Maximum drawdown	45.13%	7.85%
Skewness	−11.87	−0.98
Excess kurtosis	162.76	5.12

Studying the largest absolute monthly return reveals something important about the Equity Market Neutral index. The largest absolute monthly return is −40.46%, in November 2008, during the global financial crisis. Over the full sample, the index has a maximum drawdown of 45.13%, meaning that at its worst point an investor would have lost 45.13% from the index's previous peak.

The full sample also shows a skewness of −11.87 and an excess kurtosis of 162.76. Negative skewness means returns are lopsided to the left: most months show small gains, but there are rare, very large losses. Excess kurtosis measures how fat the tails of the distribution are compared with a normal distribution, which has an excess kurtosis of 0. A value of 162.76 is extremely high, meaning extreme returns occur far more often than a normal distribution would predict. When the November 2008 observation is removed, the statistics change dramatically: the monthly geometric average rises from 0.38% to 0.62%, the maximum drawdown falls from 45.13% to 7.85%, skewness moves from −11.87 to −0.98, and excess kurtosis drops from 162.76 to 5.12. A single month therefore drives almost all of the index's measured tail risk.

A data-quality issue seems unlikely: the dataset has no missing values, all returns are in consistent decimal form, and the extreme value falls in one of the most turbulent months in financial history. An index-construction issue is more plausible. An equity market neutral strategy balances long and short stock positions so that market movements cancel out, so it should be relatively protected during a market crash. A 40% loss in one month is therefore surprising for this strategy. Since hedge fund indices are often weighted by fund size, a severe loss at one or a few large funds could have pulled the whole index down, making the result reflect the index's composition rather than the typical market-neutral fund. Still, the loss may also reflect genuine economic tail risk. During the 2008 crisis, many hedge funds suffered from falling liquidity, forced selling and high leverage, which could affect even strategies designed to be market neutral. Moreover, even without November 2008, the index still shows negative skewness and positive excess kurtosis, so it remains exposed to occasional large losses.

In conclusion, the observation should remain in the primary analysis, since it is a real loss that investors in the index experienced, but the full-sample tail statistics should be interpreted with caution because they depend heavily on a single month.

Part III: Cross-strategy performance

5. Summary statistics

6. Performance relative to the DJCS Hedge Fund Index

Using the broad index as the benchmark for all strategies assumes they are all trying to deliver the same kind of return, which is not the case. Table 4 shows that the strategies with the largest tracking errors are those whose approach differs most from the broad hedge fund universe: Dedicated Short Bias (21.79%), which by design bets on falling prices, Managed Futures (12.70%), Emerging Markets (10.89%) and Equity Market Neutral (10.77%). For these strategies, the large deviations and negative information ratios may mainly reflect that they are designed to be different from the average hedge fund, rather than poor manager skill. For example, Dedicated Short Bias would be expected to underperform a broad index in a period when markets mostly rise. A benchmark matched to each strategy's style would give a fairer evaluation.

Table 3: Summary statistics of the nine strategy indexes (annualized; skewness and kurtosis of monthly excess returns)

Strategy	Arith. ret	Geo. ret	Vol	Sharpe	MDD	Skew	Ex. kurt
Equity Long/Short	9.04%	8.89%	9.90%	0.61	22.14%	−0.07	3.20
Equity Market Neutral	5.29%	4.68%	10.22%	0.22	45.13%	−11.99	165.33
Dedicated Short Bias	−2.66%	−4.01%	16.97%	−0.34	64.24%	0.65	1.31
Global Macro	11.51%	11.63%	9.64%	0.88	27.00%	−0.10	3.90
Managed Futures	6.13%	5.59%	11.69%	0.26	17.88%	0.02	−0.03
Emerging Markets	7.99%	7.08%	14.89%	0.33	45.33%	−0.81	5.14
Event Driven	8.83%	8.98%	6.29%	0.92	19.30%	−2.24	11.38
Convertible Arbitrage	7.35%	7.34%	6.90%	0.62	33.00%	−2.57	15.82
Fixed Income Arbitrage	5.24%	5.19%	5.73%	0.38	29.10%	−4.23	29.38

Arith. ret and Geo. ret are annualized total returns; Vol is the annualized volatility of monthly excess returns; Sharpe is the annualized Sharpe ratio; MDD is the maximum drawdown depth; Skew and Ex. kurt are the skewness and excess kurtosis of monthly excess returns.

Table 4: Active performance relative to the DJCS Hedge Fund Index (annualized)

Strategy	Active return	Tracking error	Information ratio
Equity Long/Short	0.63%	5.57%	0.11
Equity Market Neutral	−3.12%	10.77%	−0.29
Dedicated Short Bias	−11.07%	21.79%	−0.51
Global Macro	3.10%	5.69%	0.54
Managed Futures	−2.28%	12.70%	−0.18
Emerging Markets	−0.43%	10.89%	−0.04
Event Driven	0.42%	4.98%	0.08
Convertible Arbitrage	−1.06%	6.87%	−0.15
Fixed Income Arbitrage	−3.17%	6.57%	−0.48

7. Equity Long/Short, Global Macro and Convertible Arbitrage

The arithmetic return is the simple average of monthly returns, annualized. It ignores compounding, meaning that over time returns are earned on a growing (or shrinking) amount of wealth, and it says nothing about how gains and losses are distributed. Global Macro has the highest arithmetic return (11.51%), but this alone does not tell us how volatile or uneven its returns were.

The geometric return is the constant annual return that would produce the same terminal wealth, so it reflects the growth an investor actually earns. The gap between arithmetic and geometric returns grows with volatility. For these three strategies, volatility is moderate and the gap is small, so the ranking is unchanged: Global Macro first (11.63%), Long/Short Equity second (8.89%) and Convertible Arbitrage third (7.34%).

The Sharpe ratio measures excess return per unit of total volatility. Global Macro again ranks first (0.88). However, Long/Short Equity, despite its higher geometric return, has a slightly lower Sharpe ratio than Convertible Arbitrage (0.61 vs. 0.62), because its volatility is higher (9.90% vs. 6.90%). It earns more, but less per unit of risk.

Maximum drawdown measures the worst peak-to-trough loss. Here, Long/Short Equity ranks first with the smallest drawdown (22.14%), while Global Macro is second (27.00%) and Convertible Arbitrage is last (33.00%). Convertible Arbitrage shows why this measure matters: its strongly negative skewness (−2.57) and high excess kurtosis (15.82) indicate that most months are calm, which keeps its volatility low and its Sharpe ratio respectable, but occasional large losses create the deepest drawdown. The Sharpe ratio does not capture this, because volatility treats gains and losses the same and ignores tail risk. Overall, the rankings differ because each measure captures a different dimension of performance: average return, compounded growth, risk-adjusted return and worst-case loss. No single measure is enough to compare the strategies.

Part IV: Path dependence and smoothing

8. Global Macro: cumulative wealth and drawdown

Starting from 100, Global Macro's wealth grows to about 770 by June 2012, but not in a straight line. The maximum drawdown starts at the peak of July 1998, when wealth was 250.32, and reaches its lowest point in

Table 5: Comparison of the three strategies (rank in brackets; lowest drawdown ranks first)

	Arith. ret	Geo. ret	Sharpe	MDD
Equity Long/Short	9.04% (2)	8.89% (2)	0.61 (3)	22.14% (1)
Global Macro	11.51% (1)	11.63% (1)	0.88 (1)	27.00% (2)
Convertible Arbitrage	7.35% (3)	7.34% (3)	0.62 (2)	33.00% (3)

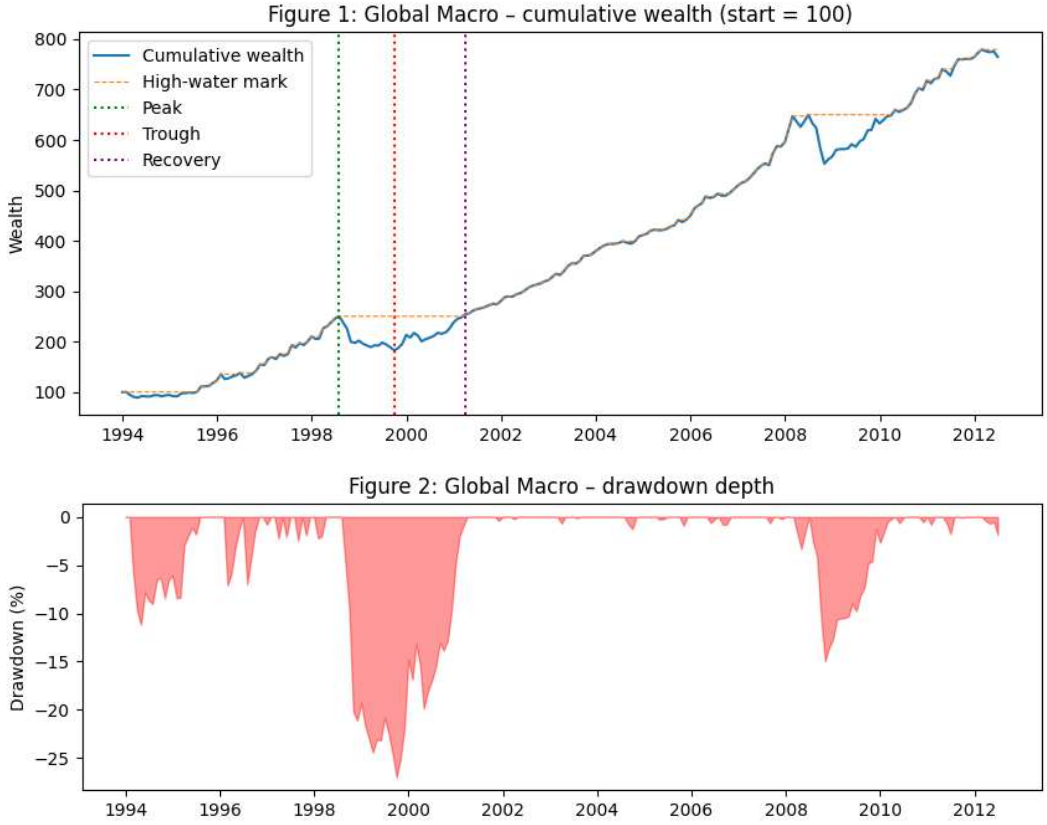


Figure 1: Global Macro. Top: cumulative wealth starting at 100, with the high-water mark and the peak, trough and recovery of the maximum drawdown. Bottom: drawdown depth in percent (shown as negative values).

September 1999, when wealth was 182.73: a loss of 27.00% from the peak. The index went back above its previous high-water mark in March 2001, so the high-water mark was recovered within the sample, but it took 32 months from the peak. The chart also shows a smaller drawdown during the 2008 financial crisis, which was recovered by 2010.

9. Lag-one autocorrelation

The three strategies with the highest positive lag-one autocorrelation are Convertible Arbitrage ($\rho_1 = 0.56$), Fixed Income Arbitrage ($\rho_1 = 0.53$) and Event Driven ($\rho_1 = 0.36$). By contrast, Managed Futures, Equity Market Neutral and Global Macro show autocorrelations close to zero.

Many hedge fund strategies hold illiquid assets that do not trade every month. When no recent market price is available, the fund values these assets using the last observed price (a stale price) or an estimated value that adjusts gradually (a smoothed valuation). As a result, a real change in value is not recorded all at once but spread over several months. For example, a true loss of 6% in one month may be reported as -2% in each of three consecutive months. Because the same shock appears in consecutive months, reported returns become positively autocorrelated: a bad month tends to be followed by another bad month.

Conventional square-root-of-time annualization (monthly volatility $\times\sqrt{12}$) assumes that monthly returns are independent. With positive autocorrelation, losses accumulate over time, so the true long-horizon risk is larger than $\sigma \times \sqrt{12}$. In addition, smoothing lowers the measured monthly volatility itself, because large shocks are split into smaller monthly moves. Both effects make the annualized volatility too low and the annualized Sharpe ratio too high.

Table 6: Global Macro: maximum drawdown

Peak date	Trough date	Depth	Recovery date
31 July 1998 (wealth 250.32)	30 September 1999 (wealth 182.73)	27.00%	30 March 2001

Table 7: Lag-one autocorrelation of monthly excess returns

Convertible Arbitrage	**0.56**	Emerging Markets	0.31	Global Macro	0.07
Fixed Income Arbitrage	**0.53**	Equity Long/Short	0.19	Equity Market Neutral	0.06
Event Driven	**0.36**	Dedicated Short Bias	0.08	Managed Futures	0.03

The results are consistent with this. The three most autocorrelated strategies also have the three lowest annualized volatilities in Table 3 (Fixed Income Arbitrage 5.73%, Event Driven 6.29%, Convertible Arbitrage 6.90%), and Event Driven has the highest Sharpe ratio of all strategies (0.92). At the same time, all three show strongly negative skewness and high excess kurtosis (for example, Fixed Income Arbitrage: skewness -4.23, excess kurtosis 29.38), and Convertible Arbitrage has a maximum drawdown of 33.00%. Their smooth returns and low volatility therefore likely understate their true long-horizon risk: a smooth return series does not necessarily mean a low-risk strategy.

10. Sharpe ratio vs. tail risk: Convertible Arbitrage

Table 8: Sharpe ratio vs. volatility and tail-risk measures (sorted by Sharpe ratio)

Strategy	Sharpe	Vol	Skew	Ex. kurt	MDD
Event Driven	0.92	6.29%	-2.24	11.38	19.30%
Global Macro	0.88	9.64%	-0.10	3.90	27.00%
Convertible Arbitrage	0.62	6.90%	-2.57	15.82	33.00%
Equity Long/Short	0.61	9.90%	-0.07	3.20	22.14%
Fixed Income Arbitrage	0.38	5.73%	-4.23	29.38	29.10%
Emerging Markets	0.33	14.89%	-0.81	5.14	45.33%
Managed Futures	0.26	11.69%	0.02	-0.03	17.88%
Equity Market Neutral	0.22	10.22%	-11.99	165.33	45.13%
Dedicated Short Bias	-0.34	16.97%	0.65	1.31	64.24%

Looking only at the Sharpe ratio, Convertible Arbitrage appears to be a strong strategy: its annualized Sharpe ratio of 0.62 is the third highest of the nine strategies, achieved with a low annualized volatility of only 6.90% (Table 8). The other measures tell a different story. Its skewness of -2.57 indicates a long left tail: most months deliver small gains, but occasionally the strategy suffers large losses. Its excess kurtosis of 15.82 indicates fat tails, meaning extreme months occur far more often than a normal distribution would predict; combined with the negative skewness, these extremes are mainly losses. This is confirmed by its maximum drawdown of 33.00%, the fourth largest in the sample, and much deeper than that of Long/Short Equity (22.14%), which has almost the same Sharpe ratio (0.61).

The Sharpe ratio misses this because it only uses average excess return and volatility. Volatility treats gains and losses symmetrically and cannot distinguish between many small fluctuations and a few severe losses. Convertible Arbitrage earns steady returns in most months, which keeps its volatility low, but it is exposed to rare, large losses. The economic risk the Sharpe ratio misses is crash or liquidity risk: the strategy holds relatively illiquid positions that perform well in calm markets but can lose heavily during periods of market stress, when liquidity dries up and positions must be sold at unfavourable prices. In addition, its lag-one autocorrelation of 0.56, the highest in the sample, suggests that smoothed valuations may understate its volatility and overstate its Sharpe ratio. An investor relying only on the Sharpe ratio would therefore underestimate how much this strategy can lose in a crisis.

Part V: Factor attribution for Equity Long/Short

Dependent variable: the Equity Long/Short monthly excess return X_t. One-factor model: $X_t = \alpha + \beta_M(\text{Mkt}-\text{RF})_t + \varepsilon_t$. Four-factor model: $X_t = \alpha + \beta_M(\text{Mkt}-\text{RF})_t + \beta_S\text{SMB}_t + \beta_V\text{HML}_t + \beta_U\text{UMD}_t + \varepsilon_t$. Both are estimated by OLS with an intercept and conventional standard errors.

11. Regression results

Table 9: Equity Long/Short: one-factor and four-factor regressions (OLS standard errors)

	One-factor	Four-factor
Alpha (monthly)	0.27%	0.16%
Alpha (annualized)	3.19%	1.93%
t(alpha)	2.10	1.69
p(alpha)	0.037	0.093
95% CI alpha, monthly ($\hat{\alpha} \pm 1.96\,SE$)	[0.02%, 0.51%]	[−0.03%, 0.35%]
95% CI alpha, annualized	[0.21%, 6.17%]	[−0.31%, 4.17%]
β_M (Mkt−RF)	0.460	0.468
β_S (SMB)		0.177
β_V (HML)		−0.083
β_U (UMD)		0.166
Adjusted R^2	0.570	0.764
Residual volatility (annualized)	6.50%	4.81%
Appraisal ratio (annualized)	0.49	0.40
Lag-one autocorrelation of residuals	0.15	0.18

In the one-factor model, the Equity Long/Short index has a positive monthly alpha of 0.27%, or 3.19% annualized, which is the average excess return not explained by market exposure. Alpha is statistically significant at the 5% level ($t = 2.10$, $p < 0.05$), and its 95% confidence interval (0.21% to 6.17% per year) excludes zero, although only barely. We can therefore reject the hypothesis that alpha is zero. The market beta of 0.46 means that when the market's excess return rises by 1%, the strategy's excess return rises by about 0.46% on average. However, the adjusted R^2 is only 0.57, so about 43% of the variation in the strategy's returns is not explained by the market. Annualized residual volatility is 6.50%, giving an appraisal ratio of 0.49.

In the four-factor model, alpha falls to 0.16% per month (1.93% annualized) and is no longer significant at the 5% level ($t = 1.69$, $p > 0.05$). Its confidence interval (−0.31% to 4.17% per year) includes zero, so we cannot rule out that the true alpha is zero. The strategy has significant positive loadings on the market (0.47), SMB (0.18) and UMD (0.17), and a significant negative loading on HML (−0.08). The adjusted R^2 rises to 0.76, so these factors explain considerably more of the return variation. Residual volatility falls to 4.81%, but alpha falls proportionally more, so the appraisal ratio declines from 0.49 to 0.40.

Conventional OLS standard errors assume independent residuals, but the residuals show positive lag-one autocorrelation (0.15 in the one-factor model and 0.18 in the four-factor model). This means OLS standard errors are likely understated, so the t-statistics are too high and the confidence intervals too narrow. Since the one-factor alpha is only just significant, we should be cautious: correcting for this dependence could make it insignificant.

12. Factor loadings in the four-factor model

Table 10: Four-factor loadings

Factor	Loading	t-stat	p-value
Mkt−RF	0.468	21.15	0.0000
SMB	0.177	6.15	0.0000
HML	−0.083	−2.70	0.0075
UMD	0.166	9.06	0.0000

All four factor loadings are statistically significant: their p-values are below 0.05, so we can reject the hypothesis that each loading is zero at the 5% level. However, statistical significance only tells us that we are confident each exposure differs from zero, not how much it matters.

- **Market** ($\beta_M = 0.47$, $t = 21.15$). When the market's excess return rises by 1%, the index's excess return rises by about 0.47% on average, and it falls by a similar amount when the market declines. The index therefore moves in the same direction as the market but only about half as much.
- **Size** ($\beta_S = 0.18$, $t = 6.15$). The positive SMB loading means the index tends to perform better when small-cap stocks outperform large-cap stocks, suggesting a tilt toward smaller companies.
- **Value** ($\beta_V = -0.08$, $t = -2.70$). The negative HML loading means the index tends to perform slightly worse

when value stocks outperform growth stocks, suggesting a mild tilt toward growth stocks.
- **Momentum** ($\beta_U = 0.17, t = 9.06$). The positive UMD loading means the index benefits when recent winners continue to outperform recent losers, suggesting managers tend to hold recent winners and short recent losers.

Together, these loadings describe a net-long equity strategy with tilts toward small-cap and momentum stocks and, to a lesser extent, growth stocks. Since the model uses returns rather than holdings, these are inferred exposures: the index behaves like a portfolio with these characteristics.

A loading can be statistically significant without being economically important. Economic importance depends on how large the loading is, that is, how much the index's return actually changes when the factor moves. The market loading is both the most significant and by far the largest, about 2.5 to 3 times the SMB and UMD loadings, so market exposure is the main driver of the strategy's returns and risk. SMB and UMD have moderate loadings (0.18 and 0.17), so they are economically meaningful secondary exposures. By contrast, the HML loading, although significant at the 1% level ($t = -2.70$, $p = 0.0075$), is small: a 1% outperformance of value over growth lowers the index's return by only about 0.08%. The growth tilt is therefore statistically detectable but economically minor; its significance mainly reflects the long sample, which allows even small effects to be estimated precisely.

13. Comparing the alphas

In the one-factor model, the Equity Long/Short index has a monthly alpha of 0.27% (3.19% annualized), statistically significant at the 5% level ($t = 2.10$), with a 95% confidence interval of 0.21% to 6.17% per year that just excludes zero. In the four-factor model, alpha falls to 0.16% per month (1.93% annualized), is no longer significant at the 5% level ($t = 1.69$), and its confidence interval (-0.31% to 4.17% per year) includes zero. Adding SMB, HML and UMD therefore removes about 1.26 percentage points per year, roughly 40% of the original alpha, while the adjusted R^2 rises from 0.57 to 0.76.

The market-only regression cannot see the strategy's exposures to size and momentum. The four-factor model shows that the index loads positively and significantly on SMB (0.18) and UMD (0.17). When these factors are left out, the returns the strategy earns from holding small-cap and momentum stocks end up in the intercept and look like manager skill. The market beta barely changes between the two models (0.46 vs. 0.47), so the drop in alpha comes from the added factors, not from a different estimate of market exposure. The small negative HML loading (-0.08) indicates a slight growth tilt, but its effect on alpha is minor.

The evidence does not support a claim of repeatable manager skill, for several reasons:

- Once common factor exposures are controlled for, alpha is not statistically significant at the 5% level, and its confidence interval includes zero, so we cannot rule out that true alpha is zero.
- Even the one-factor alpha is fragile: it is only just significant, and the residuals show positive autocorrelation (0.15 to 0.18), which means conventional OLS standard errors are likely too small.
- Showing that skill is repeatable would require evidence that alpha persists over time, for example across subperiods or out of sample, which these full-sample regressions do not test.

Most of the apparent outperformance of the Equity Long/Short index is explained by systematic exposures to the market, size and momentum. The remaining alpha is positive but statistically indistinguishable from zero, so there is no reliable evidence of repeatable manager skill.

Part VI: Practitioner memo

Global Macro shows the most consistent performance of the three: the highest geometric return (11.63%), the highest Sharpe ratio (0.88), a positive information ratio against the DJCS benchmark (0.54), limited skewness (-0.10) and low autocorrelation (0.07), suggesting its reported volatility is reliable.

Convertible Arbitrage looks attractive on volatility (6.90%), but its strongly negative skewness (-2.57) and high excess kurtosis (15.82) reveal rare but large losses, confirmed by a 33.00% maximum drawdown. Its high autocorrelation (0.56) suggests smoothed prices may understate its risk, and it underperforms the benchmark (negative active return).

For Equity Long/Short, the four-factor model shows that its apparent alpha is largely explained by market, size and momentum exposures; once these are controlled for, alpha is no longer statistically significant.

Therefore, I would recommend further due diligence on Global Macro. Its headline Sharpe ratio hides substantial drawdown risk: a 27.00% maximum drawdown, deeper than Equity Long/Short's (22.14%), with moderately fat tails (excess kurtosis 3.90). The loss lasted from July 1998 to September 1999, and the previous peak was recovered only in March 2001.

Before investing, I would also like to request:

- A return attribution by asset class and risk factor, so that performance can be tested against factors suited to a macro strategy (rates, currencies, commodities, trend), as we did for Equity Long/Short, to determine whether returns reflect genuine manager alpha.
- Performance during stress periods, including geopolitical and macroeconomic shocks, to understand how deep and prolonged losses can become.